

# MD KAMRAN CHOWDHURY SHISHER

Department of Electrical and Computer Engineering

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Google Scholar: <https://scholar.google.com/citations?user=LKaamzMAAAAJ&hl=en>

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## Education

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| <b>Ph.D. in Electrical Engineering</b><br>Auburn University, Auburn, AL<br>Advisor: Prof. Yin Sun<br><b>Thesis Title: Timely Inference over Networks</b> | Aug. 2018 – May 2024  |
| <b>M.S. in Electrical Engineering</b> M.S. in Electrical Engineering<br>Auburn University, Auburn, AL                                                    | Aug. 2022             |
| <b>B.Sc. in Electrical and Electronic Engineering</b><br>Bangladesh University of Engineering and Technology                                             | Feb. 2013 – Sep. 2017 |

## Research Interests

Communication Networks, Wireless Communications, Goal-oriented and Semantic Communication, Networked Control Systems, Machine Learning, Stochastic Optimization and Control

## Professional Experience

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| <b>Assistant Professor</b><br>Department of Electrical and Computer Engineering<br>University of Colorado Colorado Springs                                             | Aug. 2026 – Present   |
| <b>Postdoctoral Researcher</b><br>Department of Electrical and Computer Engineering, Purdue University<br>Advisors: Prof. Christopher G. Brinton and Prof. Mung Chiang | May 2024 – Aug. 2026  |
| <b>Research Assistant</b><br>Department of Electrical and Computer Engineering, Auburn University<br>Advisor: Prof. Yin Sun                                            | Aug. 2018 – May 2024  |
| <b>Teaching Assistant</b><br>Auburn University                                                                                                                         | Jan. 2020 – May 2024  |
| <b>Lecturer</b><br>Department of Electrical and Electronic Engineering<br>Bangladesh Army University of Science and Technology                                         | Nov. 2017 – June 2018 |

## Awards and Honors (selected)

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|---------------------------------------------------------------------------|------|
| <b>Auburn University Graduate School Distinguished Dissertation Award</b> | 2026 |
| ECE Nominee for Purdue CoE Outstanding Postdoc Award                      | 2026 |
| <b>The IEEE ComSoc William R. Bennett Prize</b>                           | 2025 |
| Research Profile featured in ACM SIGMETRICS Performance Evaluation Review | 2023 |
| NSF Student Travel Grant, ACM MobiHoc                                     | 2023 |
| NSF Travel Grant, North American School of Information Theory             | 2023 |

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|-------------------------------------------------|------------------|
| ACM SIGMOBILE Student Travel Grant, ACM MobiHoc | 2022             |
| IEEE INFOCOM Student Conference Grant           | 2022             |
| NSF Student Conference Award, IEEE INFOCOM      | 2021             |
| Dean's Award, BUET, Bangladesh                  | 2017, 2016, 2015 |

## Publications

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### Journal Papers: (Student Mentees Underlined)

- J7. **M. K. C. Shisher**, A. Piaseczny, Y. Sun, and C. Brinton, "Computation and Communication Co-scheduling for Multi-Task Remote Inference," under review in IEEE/ACM Transactions on Networking.
- J6. T. Z. Ornee, **M. K. C. Shisher**, C. Kam, and Y. Sun, "Remote Safety Monitoring: Status Updating for Situational Awareness Maximization," under review in IEEE Transactions on Information Theory.
- J5. C. Ari, **M. K. C. Shisher**, E. Uysal, and Y. Sun, "Goal-Oriented Status Updating for Real-time Remote Inference over Networks with Two-Way Delay," IEEE Transactions on Networking, vol. 34, pp. 4011-4025, 2026.
- J4. **M. K. C. Shisher**, V. Tripathi, C. Brinton, and M. Chiang, "AoI-based Scheduling of Correlated Sources for Timely Inference," IEEE Transactions on Networking, vol. 34, pp. 2181-2195, 2026.
- J3. **M. K. C. Shisher**, Y. Sun, and I. Hou, "Timely Communications for Remote Inference," IEEE/ACM Transactions on Networking, vol. 32, no. 5, pp. 3824-3839, 2024. [[IEEE ComSoc William R. Bennett Prize \(2025\) Winner](#)]
- J2. **M. K. C. Shisher**, B. Ji, I. Hou, and Y. Sun, "Learning and Communications Co-Design for Remote Inference Systems: Feature Length Selection and Transmission Scheduling," IEEE Journal on Selected Areas in Information Theory, vol. 4, pp. 524-538, 2023.
- J1. K. Yan, **M. K. C. Shisher**, and Y. Sun, "A Transfer Learning-Based Deep Convolutional Neural Network for Detection of Fusarium Wilt in Banana Crops," AgriEngineering, vol. 5, no. 4, pp. 2381-2394, 2023.

### Conference Papers: (Student Mentees Underlined)

- C12. Y. Zou, **M. K. C. Shisher**, V. Tripathi, and C. Brinton, "Communication-Efficient Personalized Adaptation via Federated-Local Model Merging," submitted, 2026.
- C11. Y. Zou, **M. K. C. Shisher**, V. Tripathi, and C. Brinton, "Distributed Online Convex Optimization with Interference Constraints: The Role of Gradient Freshness," submitted, 2026.
- C10. **M. K. C. Shisher**, V. Tripathi, C. Brinton, and M. Chiang "Online Learning of Whittle Indices for Restless Bandits with Non-Stationary Transition Kernels," submitted, 2026.
- C9. A. Piaseczny, **M. K. C. Shisher**, S. Wang, and C. Brinton "RCCDA: Adaptive Model Updates in the Presence of Concept Drift under a Constrained Resource Budget," NeurIPS, 2025. [[Acceptance Rate: 24.52%](#)]
- C8. **M. K. C. Shisher**, V. Tripathi, C. Brinton, and M. Chiang "AoI-based Scheduling of Correlated Sources for Timely Inference," IEEE ICC, 2025.
- C7. **M. K. C. Shisher**, A. Piaseczny, Y. Sun, and C. Brinton, "Computation and Communication Co-scheduling for Timely Multi-Task Inference at the Wireless Edge," IEEE INFOCOM, 2025. [[Acceptance Rate: 18.6%](#)]
- C6. C. Ari, **M. K. C. Shisher**, E. Uysal, and Y. Sun, "Goal-Oriented Communications for Remote Inference with Two-Way Delay," IEEE ISIT, 2024.
- C5. **M. K. C. Shisher** and Y. Sun, "On the Monotonicity of Information Aging," IEEE INFOCOM ASoI Workshop, 2024.
- C4. T. Z. Ornee, **M. K. C. Shisher**, C. Kam, and Y. Sun, "Context-aware Status Updating: Wireless Scheduling for Maximizing Situational Awareness in Safety-critical Systems," IEEE MILCOM, 2023.

- C3. **M. K. C. Shisher** and Y. Sun, “How Does Data Freshness Affect Real-time Supervised Learning?” ACM MobiHoc, 2022. [\[Acceptance Rate: 19.8%\]](#)
- C2. **M. K. C. Shisher**, H. Qin, L. Yang, F. Yan, and Y. Sun, “The Age of Correlated Features in Supervised Learning based Forecasting,” IEEE INFOCOM AoI Workshop, 2021.
- C1. **M. K. C. Shisher**, T. Z. Ornee, and M. F. Hossain, “QoS aware user association in massive MIMO enabled hetnets for DTU and NDTU traffic,” IEEE ICAEE, 2017.

## Research Experience

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- 1 **Postdoctoral Researcher** May 2024-Present  
Purdue University
  - Optimized Machine learning (ML) model training methods for **resource-constrained edge devices** [C9] and designed a communication-efficient federated fine-tuning algorithm for **Large Language Models** on edge devices [C12].
  - Developed an **online resource allocation algorithm** for restless multi-armed bandits (RMABs) in unknown, non-stationary settings [C10].
  - Developed a **distributed convex optimization** algorithm under interference constraints [C11].
  - Developed **wireless scheduling algorithms for coupled or correlated sources** under interference constraints to minimize remote inference error [J6, C8].
  - Jointly optimized **communication scheduling and feature computation** to improve the performance of multi-task remote inference systems [J7, C7].
- 2 **Research Assistant** Aug. 2018-May 2024  
Auburn University
  - Introduced a novel research topic on **Remote Inference**. Established an **information-theoretic theory** to interpret **the impact of information freshness on inference error** [C2, C3, C5] by using a generalized conditional entropy and by introducing the concept of  $\epsilon$ -Markov chain and  $\epsilon$ -**data processing inequality**.
  - Optimized the performance of remote inference systems by designing **scheduling algorithms** for both the single-user, single channel and the multi-user, multi-channel systems [J2, J3, C6, C3]. We proposed a new “**selection-from-buffer**” scheduling model to optimize the remote inference systems. Paper [J3] based on this research received [IEEE ComSoc William R. Bennett Prize \(2025\)](#).

## Teaching Experience

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1. **Guest Lecturer** Fall 2024  
ECE 547: Introduction to Computer Communication Networks  
Purdue University
2. **Teaching Assistant** Spring 2023, 2024  
ELEC-5970/6970: Applied Statistical and Machine Learning  
Auburn University
3. **Teaching Assistant** Spring 2020, 2021, 2022, Fall 2022  
ELEC-7970: Reinforcement Learning  
Auburn University
4. **Teaching Assistant** Fall 2022  
ELEC-2120: Signals and Systems  
Auburn University
5. **Lecturer** Nov 2017-June 2018  
Bangladesh Army University of Science and Technology
  - Courses: Communication Theory, Digital Signal Processing, and Telecommunication Engineering

- Labs: Communication Engineering, Digital Signal Processing

## Research Proposals

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**NSF VINES: Track 2:** NextG Smart Factory/Manufacturing, submitted Sep. 2025

- **Contributions:** Thrust. Smart Factory/Manufacturing Operational Status.
- **PIs:** H. Kwon (Wichita State U), S. Kim (Iowa State U), M. Zhang (Mississippi State U); Senior Personnel include C. Brinton (Purdue U) and others from Ericsson Research, Penn State, Arizona State U, and UT Arlington.

**NSF CPS: Medium:** Name-Based Design of Distributed Intelligence for Smarter Communities, submitted June 2025

- **Contributions:** Thrust. Fine-Grained Control: Updating Models under Heterogeneity and Dynamics
- **PIs:** E. Yeh (Northeastern U), C. Brinton (Purdue U), S. Sannigrahi (Tennessee Tech), H. Newman (Caltech).

**NSF PDaSP:** Privacy-Preserving Aggregation of Demand Flexibility for Sustainable Power Systems, submitted Sep. 2024

- **Contributions:** Thrust. Tackling Hierarchical and Heterogeneous Environments
- **PIs:** J. Qin (Purdue U), A. Hashemi (Purdue U), C. Brinton (Purdue U), K. Poolla (UC Berkeley).

## Student Mentoring

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| <b>Ananth Ram Rajagopalan</b> , Purdue University Graduate Student   | Nov 2025-Present   |
| <b>Yinan Zou</b> , Purdue University Graduate Student                | Jan 2025-Present   |
| <b>Adam Piaseczny</b> , Purdue University Graduate Student           | May 2024-Present   |
| <b>Sam Chamoun</b> , Auburn University Undergraduate Student         | Aug. 2023-May 2024 |
| <b>Zachary Gayford</b> , Auburn University Undergraduate Student     | Jan 2024-May 2024  |
| <b>Cason B. Vazquez</b> , Auburn University Undergraduate Student    | Aug 2023-Dec 2023  |
| <b>Justin Tran</b> , Auburn University Undergraduate Student         | May 2021-May 2022  |
| <b>Cagri Ari</b> , Middle East Technical University Graduate Student | Aug 2023-Present   |
| <b>Mengxue Li</b> , Tuskegee University Graduate Student             | Aug 2023-Present   |
| <b>Kevin Yan</b> , Auburn High School Student                        | Aug 2022-Dec 2023  |

## Talks

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**Computation and Communication Co-scheduling for Timely Multi-Task Inference at the Wireless Edge**

- IEEE INFOCOM, London, UK May 2025

**Timely Inference over Networks**

- PhD Dissertation Defense April 2024
- Invited Talk at Purdue University, 2024 March 2024
- Invited Talk at Northwestern University, 2024 March 2024
- Invited Poster Presentation at ITA, 2024 Feb 2024
- Invited talk at Southeast Control Conference, 2024 Feb 2024
- Talk at Dept. of ECE, Auburn University Feb 2024

**Learning and Communications Co-design For Remote Inference: Feature Length Selection and Transmission Scheduling**

- Invited Talk at University of Maryland, College Park, MD Oct 2023
- Graduate Engineering Research Showcase, Auburn University Oct 2023

## **Communications of Timely Information for Real-time Machine Learning and Networked Intelligence**

- North American School of Information Theory, Philadelphia, PA June 2023

## **How Does Data Freshness Affect Real-time Supervised Learning?**

- Information Theory Application Workshop, San Diego, CA Feb 2023
- Auburn University Research Symposium (Poster Presentation) March 2023
- ACM MobiHoc, Seoul, South Korea. Oct 2022
- College of Engineering Research Showcase at the U.S. Space and Rocket Center, Huntsville, AL (Poster Presentation) Aug 2022

## **The Age of Correlated Features in Supervised Learning based Forecasting**

- IEEE INFOCOM AOI Workshop, Vancouver, BC, Canada (virtual) May 2021

## **Services**

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### **Reviewer for Journal Manuscript Submissions**

- IEEE Transactions on Information Forensics & Security, 2025
- IEEE Transactions on Signal Processing, 2025
- IEEE Transactions on Communications, 2024, 2025
- IEEE Transactions on Networking, 2024, 2025
- IEEE Transactions on Mobile Computing, 2024, 2025
- IEEE Transactions on Network Science and Engineering, 2024
- IEEE Transactions on Green Communications and Networking, 2024
- IEEE Journal of Communications and Networks, 2020, 2023
- IEEE Journal on Selected Areas in Information Theory, 2023
- IEEE Open Journal of the Communications Society, 2023
- IEEE Transactions on Wireless Communications, 2022, 2024
- IEEE Journal on Selected Areas in Communication, 2020

### **Reviewer for Conference Manuscript Submissions**

- IEEE ITW, 2024
- IEEE WiOpt, 2024
- ACM MobiHoc ASoI Workshop, 2024
- IEEE ISIT, 2022
- IEEE INFOCOM, 2020, 2022
- IEEE WCNC, 2021, 2022
- IEEE INFOCOM AoI Workshop, 2019, 2020

### **TPC member**

- IEEE INFOCOM, 2027
- ACM MobiHoc, 2026
- IEEE WCNC, 2026
- IEEE WCNC, 2022
- IEEE WCNC, 2021

### **Conference Session Chair**

- IEEE INFOCOM ASoI Workshop, 2025
- IEEE INFOCOM DTWin Workshop, 2025

Maintainer of an online paper repository on Age of Information

Aug 2018-May 2024

Volunteer on E-Day, Auburn University, Auburn, AL

Feb 2020

Organizing Secretary, Bangladesh Student Organization, Auburn University

Aug 2022-Aug 2023

## Professional Membership

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IEEE Member

IEEE Information Theory Society Member

IEEE Communication Society Member

ACM SIGMOBILE Member